

Capital Home Medical Equipment

Customer Inquiry Automation – Practical Assessment Report

1. Solution Overview

To address common customer inquiries in a structured and scalable way, I designed a proof-of-concept customer inquiry assistant with two parts: a chatbot for common questions and a separate escalation service for questions that require human follow-up.

The chatbot answers frequently asked questions such as store hours, locations, rentals, product categories, respiratory services, and sleep-related services. Questions that cannot be answered reliably, such as exact pricing or live product availability, are escalated to a staff representative by email.

This keeps the solution practical. Common questions are answered quickly, while higher-risk or uncertain questions are passed to a human instead of being guessed.

2. How the Solution Works

The solution uses a curated FAQ knowledge base based on publicly available information from Capital Home Medical Equipment. This knowledge base was uploaded into the chatbot so it could respond to supported questions clearly and professionally.

When a customer asks a question, the chatbot checks whether it matches supported FAQ content. If it does, the chatbot returns a direct response. If the question requires live confirmation or falls outside approved knowledge, the inquiry is escalated. The escalation service then sends an email notification to a representative for follow-up.

For example, a question such as “What time do you open in Kanata?” can be answered automatically. A question such as “How much is a wheelchair?” should be escalated instead.

3. Tools and Technologies Used

The prototype uses Open WebUI for the chatbot interface, Ollama to run a local language model, and llama3.2:3b as the local model. It also uses a Markdown FAQ knowledge base for approved business content, FastAPI and Python for the escalation service, SMTP or Gmail for escalation emails, and Docker to run Open WebUI locally.

This combination was chosen because it is lightweight, practical, and easy to explain.

4. Option Analysis

Several alternatives could be used depending on cost, complexity, and long-term business needs.

The first option is the local prototype used in this assessment. Its main advantage is simplicity. It is low cost, easy to demonstrate, and suitable for a proof of concept. Its limitation is that production scaling and integration would require more setup.

A second option is Azure OpenAI. This would allow the chatbot to run on a managed cloud platform with better scalability and easier integration with services such as workflow

automation, storage, and monitoring. It would be a stronger long-term choice for production, but it would increase cost and complexity.

A third option would be another managed cloud AI platform such as Amazon Bedrock or Google Cloud Vertex AI. These platforms could also support a customer inquiry assistant and future integrations, but they add more complexity than needed for this demonstration.

For this assessment, the local prototype is the most practical choice because it demonstrates the business logic and escalation workflow without overcomplicating the solution.

5. Real Business Use

In a real business setting, this solution could reduce the time staff spend answering repetitive questions. Customers could receive quick responses for common inquiries, while staff can focus on questions that need live confirmation or case-specific support.

This approach is especially useful for store hours and location questions, rentals and product category questions, general service information, and routing pricing or stock questions for human follow-up.

The system is not intended to replace staff. Instead, it acts as a first-line support assistant that improves efficiency and provides a more consistent customer experience.

6. Maintenance and Updates

The solution can be maintained by updating the FAQ knowledge base as business information changes. For example, if store hours, contact details, or service offerings change, the knowledge file can be updated and re-uploaded.

The escalation service can also be extended over time. In a future version, it could connect to live systems for inventory checks, exact pricing, order status, and delivery scheduling.

7. Assumptions

This prototype is based on publicly available information and a curated FAQ knowledge base. It is not connected to live internal systems. Exact inventory, pricing, delivery timing, and order-specific details are therefore treated as escalation cases and are not answered automatically.

8. Conclusion

This prototype provides a practical and functional approach to customer inquiry automation. It combines a chatbot for common questions with a separate escalation process for inquiries that require human review. This design supports efficiency, keeps responses professional, and reduces the risk of giving incorrect information for questions that require live business data.